

1)

- a) An LDO is supplied by a 100 μF capacitor with an initial voltage of 50 V. The output voltage of the LDO is 5 V. If the LDO supplies a resistor of 10 Ω , how long does it take to reach a drop-out voltage of 10 V of the LDO?
- b) Plot the power dissipation of LDO pass transistor vs. time until the LDO reaches its drop-out voltage.
- c) If the junction-case thermal resistance of the LDO is 10 $^{\circ}\text{C/W}$, plot the junction temperature of the LDO vs. time.

2)

- a) A 1 mF capacitor of an electric vehicle has an initial voltage of 400 V. To avoid any electric arc during a crash event, the capacitor needs to be discharged down to 60 V within 3 s. How can you discharge the capacitor? Propose a method.
- b) If the capacitor has an ESR of 10 m Ω , what is the power dissipation of the capacitor during discharge event?

- 3) A DC power line feeds a sensitive electronic load. To reduce electromagnetic interference (EMI), a filter consisting of a common-mode choke (CMC) and two capacitors is used.

The components are connected as follows:

The common-mode choke is placed between the source and the load.

Two identical capacitors are connected from each line to ground on the load side (after the CMC).

Parameters:

Each winding of the CMC: $L = 2 \text{ mH}$

Capacitors: $C = 10 \text{ nF}$ (each)

- a) Explain qualitatively:

How the common-mode choke suppresses common-mode noise.

Why it has minimal effect on differential-mode signals.

- b) For common-mode noise, determine the equivalent inductance seen by the noise.
- c) Explain why the two capacitors behave as a parallel equivalent capacitance in common-mode operation.
- d) Calculate the cutoff frequency of the common-mode filter.